

Optimization Design of Silicon Carbide Power Module Based on Neural Network

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Abstract—Compared with Silicon materials, Silicon Carbide (SiC) devices are widely used due to their advantages of higher voltage resistance, higher temperature resistance, faster switching speed and bottom loss. Due to the low current carrying capacity of a single SiC chip, it is often necessary to apply multiple chips in parallel to improve the power level of the module. For the multi-chip parallel module, the size of Direct Bonded Copper (DBC) and the layout of the chip have great influence on the parasitic inductance and thermal performance of the module, so it is necessary to optimize the electro-thermal performance of the module. At present, multi-chip SiC power module can be characterized by the finite element simulation software of multi-physical field, and then the multi-chip power package design. However, the optimization design of module package by finite element simulation software needs a lot of parametric simulation, which costs much simulation time and reduces the design efficiency. Therefore, an automatic design method is urgently needed to accelerate the design cycle of the SiC power module, reduce the inductance, improve the heat dissipation efficiency and give full play to the performance of the SiC device. Firstly, taking a multi-chip parallel SiC Power Module as an example, the parasitic inductance and temperature distribution of the module were simulated and analyzed based on ANSYS Q3D and COMSOL. Secondly, Artificial Neural Network (ANN) is trained with DBC size and chip spacing as input variables, parasitic inductance and maximum junction temperature as output data, and the relationship between input and output is obtained. That is, the inductance and junction temperature of the module can be quickly obtained by directly inputting variables such as size. Finally, the root mean square error is less than 2 when comparing the finite element simulation results with the ANN training results under the same variables, which verifies the validity of the method.

Keywords- ANN, SiC Power Module, electro-thermal optimization design

I. INTRODUCTION

Power modules are widely used in electric vehicles, and the uncertain operating conditions of electric vehicles require high reliability of power modules. SiC has been widely used in the

field of electric vehicle because of its advantages of high speed and high temperature [1]. However, the chemical properties of Si are different from those of SiC, so the traditional modular design of Si materials cannot be applied to SiC materials, so it needs to be restudied. A single SiC chip has a limited load current, so it is necessary to use multiple chips in parallel in high power applications. The structure of multi-chip parallel modules is more complex, and the problem of layout is more important. Unreasonable layout will lead to the inconsistency of parasitic parameters of each parallel branch, and then affect the balanced distribution of current among chips. The parasitic inductors mainly come from the power supply terminal of power module, copper conductor on DBC and chip bonding wire. In addition, the temperature distribution is affected by the position relationship between chips, and the unreasonable chip spacing will lead to serious thermal coupling effect between chips, thus reducing the reliability of power modules [2]. As the size of power module becomes smaller and the power density increases gradually, the thermal coupling effect between chips in the module becomes stronger, so the layout design of power module becomes more important [3]. The DBC size and chip layout in power module will affect the parasitic inductance and thermal distribution, so it is necessary to carry out electro-thermal optimization design of power module. The optimal layout is based on a large number of finite element simulations, which are time-consuming and difficult to build parameterized simulation models, it is difficult to optimize multi-chip power modules efficiently by manual experience and manual design method because of the unfriendly software interface between electrical and thermal simulation. Artificial Neural Network (ANN) can be regarded as a general function approximator, that is, it can set arbitrary precision to approximate the relationship between input and output data [4]. ANN is an emerging algorithm that, like the biological brain, uses a set of signal-processing units (artificial neurons) to receive, process, and transmit information from neurons, the weights of the connections between neurons are obtained by training parameters, and the ANN is trained by the provided data sets to approximate the relationship between input and output data [5]. In the design of power module, if we

get several sets of design parameters and performance indexes and train ANN to get the relationship between parameters, we can get the simulation results of different data at the fastest speed.

In reference [6], genetic algorithm is used to design the layout of power modules with two chips and three terminals, but genetic algorithm needs a large number of population and takes up a lot of computing resources, and it is not suitable for multi-chip power module. The paper [7] improves the paper [6] by using the nesting method of genetic algorithm and enumeration, which reduces the number of population and the computational difficulty, but only considers the parasitic parameters of the module, and the efficiency of genetic algorithm is still limited. A multi-objective optimization tool, PowerSynth, was created in reference [8] to calculate the electro-thermal coupling of layout through a reduced-order model, the optimal trajectory layout and die placement path can be achieved more quickly than traditional finite element analysis (FEA), but the number of layout options is still limited. Literature [9] presents a VLSI-based placement and routing method, which provides a more diverse starting point layout for PowerSynth optimization through a synthetic circuit network table. In the layout optimization of SiC multi-chip power modules, it is necessary to reduce the parasitic inductance and the highest junction temperature, is to focus on the need to solve the problem. With the increase of power density, multi-chip becomes the trend. However, based on the layout design method in the previous literature, due to the complexity of integrated electrical and thermal performance, the parasitic parameters, maximum junction temperature, DBC size and chip coordinates are not simple linear relations, and the finite element simulation modeling is difficult, simulation time-consuming, difficult to be used in complex multi-chip power modules. In this paper, an electro-thermal co-design method for SiC multi-chip power modules is proposed. By training ANN, the mapping between input and output parameters is obtained.

In the second section, the basic principle of neural network is introduced, and in the third section, a half-bridge power module is taken as an example to analyze its electrothermal performance. In the fourth section, the ANN is trained by parameterized simulation, and the corresponding simulation value is output directly by input specified data, and verified by finite element simulation. Finally, the electro-thermal cooperative design of multi-chip Power Modules is summarized.

II. THE BASIC PRINCIPLE OF NEURAL NETWORK

Various types of artificial neural networks have been proposed in literature [10]. The particular network choice depends largely on the nature of the relationship between input and output in the data. In this paper, the relationship between chip size and electro-thermal performance is static. Therefore, we choose the forward neural network as the case study.

The artificial neural network is composed of input layer, hidden layer and output layer. The input layer can be input data or multiple layers, and the hidden layer can be multiple layers, each layer composed of neurons, the number of neurons can be arbitrarily set, used to process the data of the previous layer, as

shown in figure 1. The output of one neuron in one layer needs to be multiplied by the weight w of the output of all neurons in the previous layer, plus biasing b , to be the input of the next layer [11]. The flow is shown in figure 2.

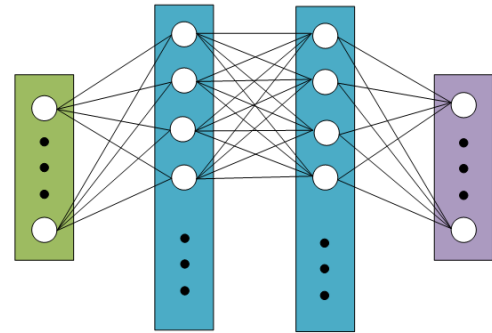


Figure 1. Structure of artificial neural network

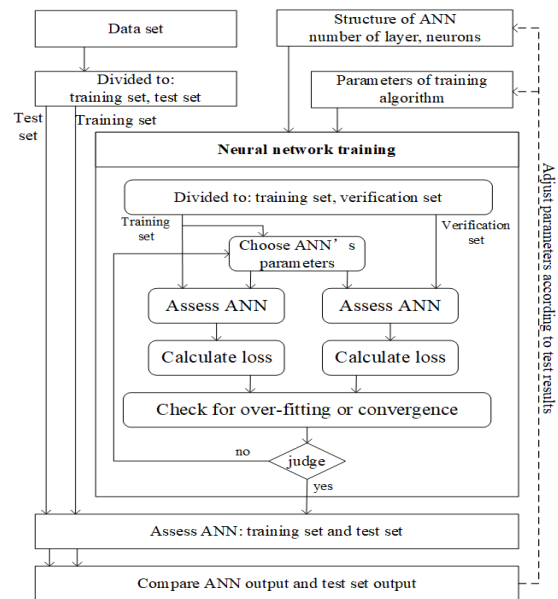


Figure 2. Neural network training flow chart

1) In order to prevent the data out of the training set from being misquoted, all the data are normalized. The data is divided into training set and test set, and the partition ratio can be arbitrary. Usually, the ratio is 80% training set and 20% test set, and the test set is used to test the accuracy of ANN;

2) The training set is further divided into training subset and verification subset, and the split proportion can be arbitrary. The proportion of this paper is 80% training subset and 20% verification subset;

3) Setting the weight and bias parameters of the neural network. The weight w and the bias parameter b of the neural network are selected randomly first, and are updated and modified according to the returned error in the course of the later iteration;

4) The training condition is evaluated by training subset and validation subset, and the error between the data and the output

of neural network is calculated by error function, which usually chooses root mean square function;

5) Comparing the error of training subset and verifying subset to judge whether ANN has entered the state of overfitting;

6) After the ANN training is finished, the data from the test set is input to the trained ANN, the output data is compared with the actual data, and the accuracy of ANN is evaluated by error function;

7) If the accuracy of the assessment is poor, then adjust the number of hidden layers and the number of neurons in ANN, and repeat the above steps.

Over-fitting is a phenomenon that artificial neural network is over-fitting to training subset samples, which results in unfavorable prediction to other samples. After a certain number of iterations, the artificial neural network is trying to adapt to the sampling error in the training data, so it is necessary to check the over-fitting phenomenon by a subset and stop training the iterated neural network.

III. MULTIPHYSICAL FIELD SIMULATION OF POWER MODULE

In this section, taking a half-bridge power module as an example, through Q3D simulation and COMSOL simulation, the parasitic inductance and maximum junction temperature are obtained. The power module is shown in figure 3(2). The half-bridge module consists of 6 chips, which are divided into 3 top-bridge-arm chips and 3 bottom-bridge-arm chips. The DC positive input is connected to the top bridge arm chip by the top DBC, and then connected to the bottom DBC and AC outputs by the gasket, and the DC negative input is connected to the bottom bridge arm chip by the bottom DBC, connect to the top DBC and AC outputs via the support. Because of the close distance between the positive and negative ends of DC, the TC plate is established for isolation. Balance stress DBC and four corner support were added in both top and bottom DBC. DC input DC voltage, AC output AC half-wave voltage. The circuit is as shown in the figure 3(1). The DBCs of module structure is shown in the figure 3 (3) and (4).

A. Power Module Parasitic Inductance Extraction

The parasitic inductance of the package module is extracted in ANSYS Q3D. In order to get the parasitic inductance value of the whole module, it is necessary to set the positive and negative DC input terminals in a network, so the chip and support are all set to copper. The material settings and parameters of each part of the half bridge module are shown in table 1. Set the DC positive input side to source and the DC negative input side to sink. The overall parasitic inductance of the module is calculated to be about 7.54nH.

TABLE I. MATERIAL SETTINGS

Components	Materials
DBC top and bottom, drive pin	Copper
DBC middle layer, TC	Al2O3
Solder layer	Solder
Chip	Si
Support	Mo.

B. Thermal Simulation of Power Module

The thermal simulation analysis of the half-bridge module is carried out by using COMSOL software. Material settings and properties are shown in table 1. Set the chip as heat source, and the heat source power consumption as a rectangular wave of which peak value is 100W, as shown in figure 4(1). To simulate the transient thermal, set the simulation time for 24s, and the calculation step size is 1. The simulation results show that the maximum junction temperature of the module is 218°C at 16s, as shown in figure 4(2), which is close to the normal critical temperature of SiC power module.

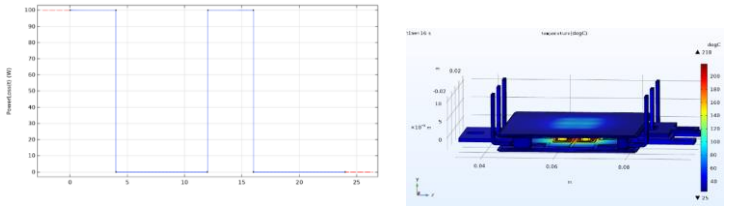


Figure 4. (1) rectangle wave of heat source power (2) temperature distribution of module

IV. MULTI-CHIP POWER MODULE ELECTROTHERMAL COOPERATIVE DESIGN BASED ON ANN

Ansys Q3D can extract parasitic inductance and COMSOL can simulate the highest junction temperature of the module. Both of them are finite element analysis software, so the accuracy is high, but the finite element analysis takes a long time. In order to get more data, the time of finite element analysis is too long, which has a great influence on the efficiency of power module layout design. In order to get more accurate relationship between DBC size, chip spacing, parasitic inductance and maximum junction temperature, the step size of parametric simulation should be as small as possible, which will lead to low simulation efficiency. Therefore, the method of training artificial neural network is used that will get the target value quickly.

For parasitic inductors of power modules, the relationship between DBC size and parasitic inductance is studied in literature [13], that is, the larger the width of DBC in the

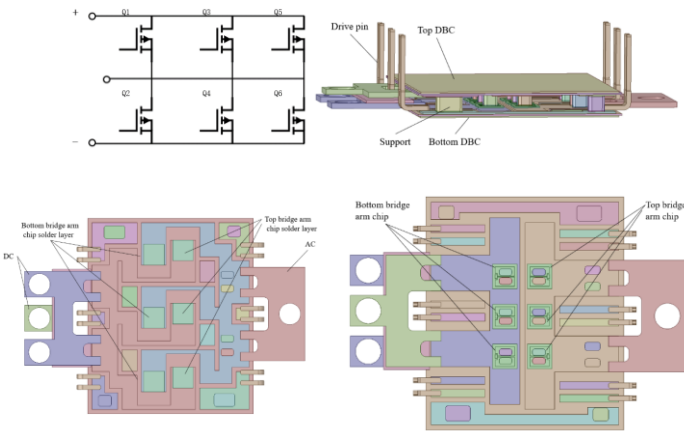


Figure 3. (1) circuit diagram of half-bridge module (2) half-bridge power module (3) top DBC (4) bottom DBC and chips

direction of current inflow, the smaller the value of parasitic inductance will be. However, for multi-chip power modules, the DBC layout is relatively complex, and there are more current paths in DBC. For the highest junction temperature, the smaller the chip spacing is, the stronger the thermal coupling between the chips is, the higher the maximum junction temperature of the chip will be, and the larger the temperature difference between the chip and other locations will be, resulting in uneven stress distribution in the module, therefore, the aging of the module is accelerated and the service life of the module is greatly reduced. However, the complexity of multi-chip parallel modules means that the relationship between chip spacing and maximum junction temperature is not a simple linear relationship. Therefore, for parasitic inductors and maximum junction temperature, we can use the training ANN method, directly input the DBC size and spacing value, output the corresponding parasitic inductor value and maximum junction temperature value.

A.Design of Power Module for Reducing Parasitic Inductance Based on ANN

Ansys Q3D software is used to parameterize the width of 6 top DBC arms and 2 bottom DBC arms in the half-bridge module. A total of $27 = 128$ sets of data can be obtained. The specific data size s are shown in table 2.

TABLE II. Q3D PARAMETERIZED VALUES

Parameter	Change value(mm)
6 top DBC arms width	1, 3
bottom DBC width	7.9, 8.8

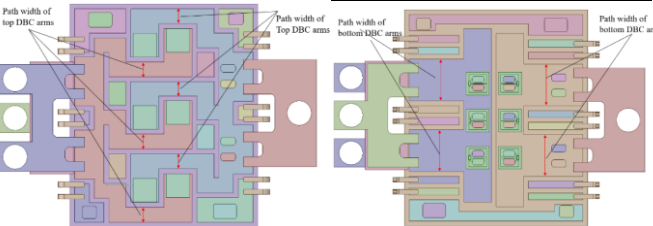


Figure 5. The corresponding parameters in the top and bottom DBC of model

According to the parameterized simulation, ANN1 is defined and its structure is determined. The 7 inputs are the width of the top DBC6 bridge arms, the width of the bottom DBC, and the output is the value of the whole parasitic inductance of the module, the network parameters of ANN1 determined by adjusting parameters are:, 13 neurons in 1 hidden layer, 7 in input layers, 1 in output layer, as shown in figure 6(1).

According to the simulation data, it can be seen that the width of the top DBC arm is negatively correlated with the parasitic inductance of the whole module, and the width of the bottom DBC is negatively correlated with the parasitic inductance of the whole module, the two width values are required to be as large as possible, while the other dimensions in the module are fixed, in order not to affect the normal operation of other parts, and not to affect other performance such as stress, according to these conclusions and constraints, specify the top and bottom DBC arm widths of 3mm and 8.8 mm, respectively. Enter ANN1 and the output is 6.7073784nH.

The modified model was verified by Q3D simulation. The overall parasitic inductance of the modified model was about 6.71nH, and the training error of ANN1 was about 0.0074nH, and the error percentage was 0.11%. The parasitic inductance of the module decreased by 0.84nH, and the loss is about 11.14%. The loss chart is shown in figure 6(2).

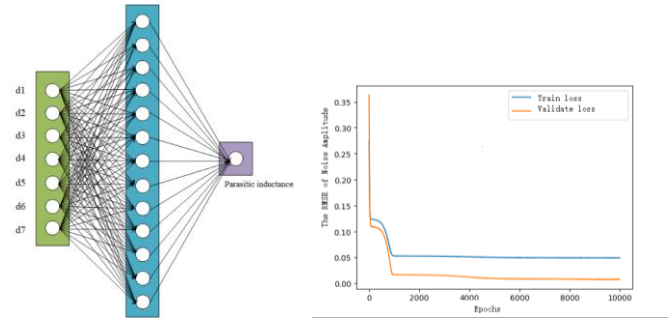


Figure 6. (1) structure of ANN1 (2) ANN1 training times and error

B.Design of Power Module Based on ANN to Reduce the Maximum Junction Temperature

The chip spacing of the half-bridge module is simulated by COMSOL software. The material setting is the same as Ansys Q3D inductor simulation. Because of the complex structure of the module, it is very difficult to carry out parametric modeling in COMSOL. This simplifies the model by replacing the inner DBC with a large rectangle, removing the shim that is not connected to the chip, and retaining the square shim, as shown in figure 7. In the parameterized setting, the chip 1-y coordinates, the chip 3-y coordinates, the chip power consumption and thermal convection coefficient four variable parameters, the other four chips according to the chip 1,3 chip position changes. As shown in table 3, a total of $3 * 3 * 5 * 3 = 135$ data sets can be obtained. In order to simplify the simulation, the steady-state solution is adopted in the parameterized simulation.

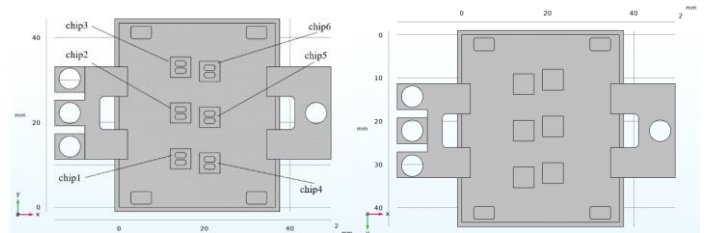


Figure 7. (1) bottom DBC and chips (2) top DBC

TABLE III. COMSOL PARAMETERIZED VALUES

Parameter	Values
Chip 1-y coordinates (mm)	5,7,9
Chip 3-y coordinates (mm)	34.6,32.6,30.6
Chip power consumption (W)	60,70,80,90,100
Thermal convection coefficient	4,000,5,000,6,000

According to the parameterized simulation of COMSOL, ANN2 is defined and its structure is determined. Four inputs are chip 1-y, chip 3-y coordinates, chip power consumption, thermal convection coefficient, and one output is the highest junction temperature of the module, the network parameters of

ANN2 determined by adjusting parameters are: 13 neurons in one hidden layer, 4 in input layer, 1 in output layer, as shown in figure 8(1). And the training and test loss is shown in figure 8(2).

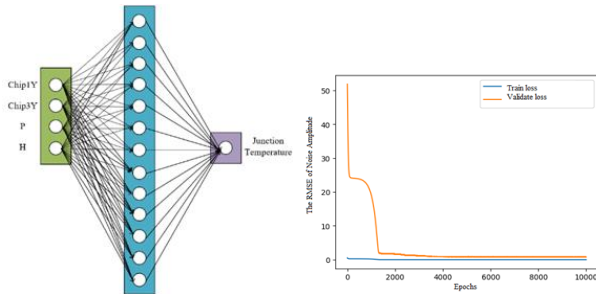


Figure 8. (1) structure of ANN2 (2) ANN2 training times and error

According to the simulation results, the 1-y coordinates and 3-y coordinates of the chips are 6mm and 35.5 mm, respectively. The power consumption of each chip is the rectangle wave in figure 4(1), and the thermal convection coefficient is 5000. The simulation results are returned to COMSOL for verification, the maximum junction temperature is 169°C, which is 49°C lower than before, as shown in figure 9.

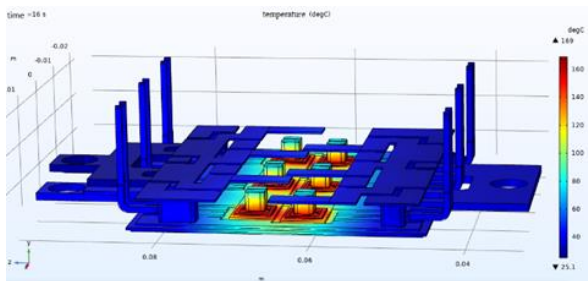


Figure 9. Maximum junction temperature simulation results of the optimized module

V. Conclusions

In order to solve the problems of modeling and simulation time-consuming in the layout optimization of SiC multi-chip

power modules, this paper proposes a training ANN to predict the simulation results under different size parameters, thus greatly reducing the simulation time, simplify the layout design of power modules. Through a concrete half-bridge power module, this paper analyzes how to use simulation data to train ANN to complete the electrothermal cooperative design of the module. The simulation results show that the parasitic inductor and the maximum junction temperature can be reduced, and the difficulty of modeling and time-consuming simulation is solved.

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